

The Program in Action

When you run *royalgameofur.py*, the output will look like this:

```
The Royal Game of Ur, by Al Sweigart
--snip--

          XXXXXXXX          .....
          Home              Goal
            v              ^
+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
|*****|   |   |   |   |   |*****|   |   |   |   |
|*  *<  <  <  <  <  |*  *<  <  <  <  <  |
|***h|   g|   f|   e|   |***t|   s|   |
+---v---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
|   |   |   |   |*****|   |   |   |   |   |   |   |
|   |   |   |   |>*  *>  |   |   |   |   |   |   |
|   i|   j|   k|***l|   m|   n|   o|   p|   |
+---^---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
|*****|   |   |   |   |*****|   |   |   |   |   |
|*  *<  <  <  <  <  |*  *<  <  <  <  <  |
|***d|   c|   b|   a|   |***r|   q|   |
+---^---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
            ^              v
          Home              Goal
        0000000          .....
```

```
It is 0's turn. Press Enter to flip...
Flips: H-H-H-H  Select token to move 4 spaces: home quit
> home
0 landed on a flower space and gets to go again.
Press Enter to continue...
--snip--
```

How It Works

Like in Project 43, “Mancala,” the spaces on the ASCII-art game board are labeled with the letters *a* through *t*. After rolling, the player can select a space that contains their token to move that token, or they can select home to begin moving a token from their home to the board. The program represents the board as a dictionary with the keys ‘a’ to ‘t’ and the values of ‘X’ and ‘0’ for the tokens (or ‘ ’ for a blank space).

Additionally, this dictionary has keys ‘x_home’, ‘o_home’, ‘x_goal’, and ‘o_goal’, and the values for these keys are seven-character strings that represent how full the homes and goals are. The ‘X’ or ‘0’ characters in these strings represent tokens at the home or goal, and ‘.’ represents an empty slot. The `displayBoard()` function displays these seven-character strings on the screen.

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1. “”The Royal Game of Ur, by Al Sweigart al@inventwithpython.com
 2. A 5,000 year old board game from Mesopotamia. Two players knock each other back as they race for the goal.
 3. More info https://en.wikipedia.org/wiki/Royal_Game_of_Ur
 4. View this code at <https://nostarch.com/big-book-small-python-projects>